

Modeling the Evolution of Wealth: A Fokker–Planck PDE Approach to Portfolio Risk

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Abstract

Portfolio risk is most often summarised by the first two moments of terminal wealth, yet the quantities an investor actually cares about—the probability of reaching a financial goal, the chance of ending below the initial capital, and the severity of the worst outcomes—are properties of the *entire* terminal distribution. We treat the distribution itself as the object of study and evolve it forward in time with the **Fokker–Planck (forward Kolmogorov) equation**, a parabolic partial differential equation whose drift encodes expected return and whose diffusion encodes volatility. For wealth following geometric Brownian motion the equation admits a closed-form log-normal solution, which we exploit as ground truth. We implement two finite-difference solvers on the log-transformed, constant-coefficient equation—an explicit forward-time/centred-space scheme and a Crank–Nicolson scheme—establish their stability by von Neumann analysis, and verify second-order spatial convergence (observed orders 2.00 and 2.15). The solver reproduces the analytic density to a maximum pointwise error of 1.8×10^{-5} , conserves probability to six digits, and agrees with a 2×10^5 -path Monte-Carlo simulation. From the propagated terminal density we recover goal-attainment probabilities, Value-at-Risk and Expected Shortfall by quadrature, matching the closed form to four decimal places, and we use these to compare conservative, balanced and aggressive equity/cash allocations on a common footing. The framework requires no closed form and therefore extends directly to state- or time-dependent coefficients where none exists.

1 Introduction

A standard description of investment risk reduces the random terminal wealth W_T to its mean and variance. This is the language of mean–variance optimisation, and it is adequate when returns are (log-)normal and preferences are quadratic. But an investor planning over a decade rarely asks “what is my variance?”. The operative questions are: *What is the probability I double my capital? What is the probability I end up poorer than I started? If things go badly, how badly?* Each of these is a functional of the full probability density $p(W, t)$, not of its moments alone.

The density of a diffusion is governed by a partial differential equation. If wealth evolves according to a stochastic differential equation (SDE), its transition density solves the *Fokker–Planck equation*, also called the forward Kolmogorov equation. Drift in the SDE becomes an advection (transport) term; diffusion becomes a second-order term. Solving this parabolic PDE forward in time propagates an initial, sharply concentrated wealth state into a spread-out terminal distribution, from which every risk metric of interest follows.

This report carries out that programme for the canonical model of portfolio wealth, geometric Brownian motion (GBM). The choice is deliberate: GBM is the rare case in which the Fokker–Planck equation can be solved in closed form—the terminal wealth is exactly log-normal—so we have an unambiguous benchmark against which to validate a numerical solver. Once validated, the same solver applies unchanged to settings with no analytic solution. Our contributions are:

1. a self-contained derivation of the Fokker–Planck equation for GBM from Itô calculus, and its reduction to a constant-coefficient advection–diffusion equation by a logarithmic change of variables (Section 2);

2. two finite-difference schemes—explicit FTCS and Crank–Nicolson—with full von Neumann stability analysis and truncation-error orders (Section 3);
3. a validation suite: closed-form comparison, grid-refinement convergence study, and Monte-Carlo cross-check (Section 4);
4. extraction of goal-attainment probability, Value-at-Risk and Expected Shortfall from the propagated density, and a comparison of asset allocations (Sections 6–7).

2 Mathematical framework

2.1 The wealth process

Let W_t denote the value at time t of a portfolio that earns an expected (arithmetic) return μ per unit time with volatility σ . The standard continuous-time model is geometric Brownian motion,

$$dW_t = \mu W_t dt + \sigma W_t dB_t, \quad W_0 > 0 \text{ given}, \quad (1)$$

where B_t is a standard Brownian motion. Returns scale with the level of wealth, which keeps $W_t > 0$ and makes proportional (percentage) gains the natural unit.

2.2 The Fokker–Planck equation

For a general one-dimensional Itô diffusion $dX_t = a(X_t) dt + b(X_t) dB_t$, the probability density $p(x, t)$ of X_t evolves according to the forward Kolmogorov / Fokker–Planck equation. We derive it in the weak (test-function) form, which is both rigorous and the form most useful for understanding the numerical scheme.

Proposition 1 (Fokker–Planck equation). *Let $dX_t = a(X_t) dt + b(X_t) dB_t$ and let $p(x, t)$ be the density of X_t . Then, for x in the interior of the state space,*

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}[a(x)p] + \frac{1}{2} \frac{\partial^2}{\partial x^2}[b^2(x)p]. \quad (2)$$

Proof. Let $\varphi \in C_c^\infty(\mathbb{R})$ be an arbitrary smooth test function with compact support. By Itô’s lemma applied to $\varphi(X_t)$,

$$d\varphi(X_t) = \left[a(X_t)\varphi'(X_t) + \frac{1}{2}b^2(X_t)\varphi''(X_t) \right] dt + b(X_t)\varphi'(X_t) dB_t.$$

Taking expectations removes the martingale (stochastic-integral) term, and dividing by dt gives the weak evolution

$$\frac{d}{dt} \mathbb{E}[\varphi(X_t)] = \mathbb{E}\left[a\varphi' + \frac{1}{2}b^2\varphi'' \right] = \int_{\mathbb{R}} \left(a(x)\varphi'(x) + \frac{1}{2}b^2(x)\varphi''(x) \right) p(x, t) dx.$$

On the other hand $\frac{d}{dt} \mathbb{E}[\varphi(X_t)] = \int_{\mathbb{R}} \varphi(x) \partial_t p(x, t) dx$. Integrating the right-hand side by parts—once for the first term, twice for the second, with all boundary terms vanishing because φ has compact support—yields

$$\int_{\mathbb{R}} \varphi \partial_t p dx = \int_{\mathbb{R}} \varphi \left(-\partial_x[a p] + \frac{1}{2} \partial_x^2[b^2 p] \right) dx.$$

Since this holds for every $\varphi \in C_c^\infty(\mathbb{R})$, the integrands agree pointwise, which is (2). $\square$

Specialising to GBM (1), where $a(W) = \mu W$ and $b(W) = \sigma W$, gives the wealth density equation

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial W}[\mu W p] + \frac{1}{2} \frac{\partial^2}{\partial W^2}[\sigma^2 W^2 p]. \quad (3)$$

Equation (3) has *variable coefficients* (μW and $\sigma^2 W^2$) and *degenerates* at $W = 0$, where both the drift and the diffusion vanish. Discretising it directly on a wealth grid is awkward: the cell-Péclet number grows with W and the degeneracy at the origin is delicate. A change of variables removes both difficulties.

2.3 Logarithmic transformation

Let $Y_t = \ln W_t$. By Itô's lemma, with $f(W) = \ln W$, $f' = 1/W$, $f'' = -1/W^2$,

$$dY_t = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dB_t = \nu dt + \sigma dB_t, \quad \nu := \mu - \frac{1}{2}\sigma^2. \quad (4)$$

Thus log-wealth is *arithmetic* Brownian motion with constant drift ν (the “Itô-corrected” or geometric growth rate) and constant diffusion σ . By Proposition 1 its density $q(y, t)$ obeys the **constant-coefficient** advection–diffusion equation

$$\boxed{\frac{\partial q}{\partial t} = -\nu \frac{\partial q}{\partial y} + \frac{1}{2}\sigma^2 \frac{\partial^2 q}{\partial y^2}} \quad y \in \mathbb{R}. \quad (5)$$

This is the equation we solve numerically. The densities are related by the change-of-variables formula: since $W = e^y$ and $|dy/dW| = 1/W$,

$$p(W, t) = \frac{1}{W} q(\ln W, t). \quad (6)$$

2.4 Closed-form solution and risk metrics

Integrating (4) gives $Y_t = \ln W_0 + \nu t + \sigma B_t$, so Y_t is exactly Gaussian and W_t is log-normal:

$$Y_t \sim \mathcal{N}\left(\underbrace{\ln W_0 + \nu t}_{m(t)}, \underbrace{\sigma^2 t}_{s(t)^2}\right), \quad q(y, t) = \frac{1}{\sqrt{2\pi} s(t)} \exp\left(-\frac{(y - m(t))^2}{2s(t)^2}\right). \quad (7)$$

Writing $m = m(T) = \ln W_0 + \nu T$ and $s = s(T) = \sigma\sqrt{T}$ for the terminal horizon, and letting Φ and Φ^{-1} be the standard-normal CDF and quantile, the risk metrics used below all have closed forms. For a goal level G and a floor level F ,

$$\mathbb{P}(W_T \geq G) = \Phi\left(\frac{m - \ln G}{s}\right), \quad \mathbb{P}(W_T \leq F) = \Phi\left(\frac{\ln F - m}{s}\right), \quad (8)$$

$$\text{VaR}_\alpha = \exp(m + s z_\alpha), \quad z_\alpha = \Phi^{-1}(\alpha), \quad \text{ES}_\alpha = \frac{1}{\alpha} e^{m + \frac{1}{2}s^2} \Phi(z_\alpha - s). \quad (9)$$

Here VaR_α is the α -quantile of terminal wealth (the level the portfolio falls below with probability α) and $\text{ES}_\alpha = \mathbb{E}[W_T \mid W_T \leq \text{VaR}_\alpha]$ is the average wealth in that lower tail. The expected-shortfall identity is the standard partial expectation of a log-normal. As a consistency check, the mean is $\mathbb{E}[W_T] = e^{m+s^2/2} = W_0 e^{\mu T}$, recovering the arithmetic growth rate.

Remark 1. The existence of (7)–(9) is exactly why GBM is the right test problem: it lets us measure the solver's error against truth. Replacing the constants μ, σ by functions $\mu(W, t), \sigma(W, t)$ —local-volatility or mean-reverting dynamics—destroys the closed form but leaves the PDE and the solver intact.

3 Numerical method

We solve (5) on a bounded interval $[y_{\min}, y_{\max}]$ with a uniform grid

$$y_j = y_{\min} + j \Delta y, \quad j = 0, \dots, N-1, \quad t^n = t_0 + n \Delta t,$$

and write $q_j^n \approx q(y_j, t^n)$. The domain is centred on the terminal mean and made wide enough, $y_{\min}/y_{\max} = m(T) \mp 7\sigma\sqrt{T}$, that the truncated mass $\Phi(-7) \approx 1.3 \times 10^{-12}$ is negligible; we impose homogeneous **Dirichlet** conditions $q_0^n = q_{N-1}^n = 0$.

Initial condition. The exact initial state is a Dirac mass $\delta(W - W_0)$, which no grid can resolve. We regularise it by starting the march at a small time $t_0 > 0$ from the exact Gaussian (7), which is smooth and well-resolved. This introduces no modelling error—it is the true solution at t_0 —and we report all errors against the exact solution at the *terminal* time, so the regularisation does not flatter the scheme.

Introduce the dimensionless *diffusion number* and *advection (Courant) number*

$$s_d = \frac{\frac{1}{2}\sigma^2 \Delta t}{\Delta y^2}, \quad s_a = \frac{\nu \Delta t}{\Delta y}. \quad (10)$$

3.1 Explicit scheme (FTCS)

Forward difference in time and centred differences in space give the explicit forward-time/centred-space (FTCS) update

$$q_j^{n+1} = q_j^n + s_d(q_{j+1}^n - 2q_j^n + q_{j-1}^n) - \frac{1}{2}s_a(q_{j+1}^n - q_{j-1}^n). \quad (11)$$

Its local truncation error is $\mathcal{O}(\Delta t) + \mathcal{O}(\Delta y^2)$: first order in time, second order in space.

Proposition 2 (von Neumann stability of FTCS). *Scheme (11) is stable in the L^2 sense if and only if*

$$s_d \leq \frac{1}{2} \quad \text{and} \quad s_a^2 \leq 2s_d. \quad (12)$$

Proof. Insert the Fourier mode $q_j^n = \xi^n e^{ikj\Delta y}$ into (11) and write $\theta = k\Delta y$. Using $e^{i\theta} - 2 + e^{-i\theta} = -4\sin^2(\theta/2)$ and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$, the amplification factor is

$$\xi(\theta) = 1 - 4s_d \sin^2(\theta/2) - i s_a \sin\theta.$$

Stability requires $|\xi|^2 \leq 1$ for all θ . With $u = \sin^2(\theta/2) \in [0, 1]$ and $\sin^2\theta = 4u(1-u)$,

$$|\xi|^2 - 1 = (1 - 4s_d u)^2 + 4s_a^2 u(1-u) - 1 = 4u \left[(4s_d^2 - s_a^2)u + (s_a^2 - 2s_d) \right].$$

The bracket is affine in u , so its maximum over $u \in [0, 1]$ occurs at an endpoint; requiring $|\xi|^2 \leq 1$ at $u \rightarrow 0$ and $u = 1$ gives, respectively, $s_a^2 \leq 2s_d$ and $s_d \leq \frac{1}{2}$, which is (12). $\square$

The condition $s_d \leq \frac{1}{2}$ is the familiar diffusive CFL limit and is the binding one here; $s_a^2 \leq 2s_d$ reduces to the mild requirement $\Delta t \leq \sigma^2/\nu^2 \approx 8$ for our parameters. The practical cost of (12) is severe: halving Δy forces Δt to drop by a factor of four.

3.2 Crank–Nicolson scheme

Averaging the explicit and implicit (backward-time) discretisations gives the Crank–Nicolson scheme, which advances $q^n \rightarrow q^{n+1}$ by solving the tridiagonal system

$$\underbrace{-\left(\frac{s_d}{2} + \frac{s_a}{4}\right) q_{j-1}^{n+1}}_{\ell} + \underbrace{(1 + s_d) q_j^{n+1}}_d - \underbrace{\left(\frac{s_d}{2} - \frac{s_a}{4}\right) q_{j+1}^{n+1}}_u = \left(\frac{s_d}{2} + \frac{s_a}{4}\right) q_{j-1}^n + (1 - s_d) q_j^n + \left(\frac{s_d}{2} - \frac{s_a}{4}\right) q_{j+1}^n. \quad (13)$$

Compactly, $\mathbf{A} \mathbf{q}^{n+1} = \mathbf{B} \mathbf{q}^n$ on the $M = N - 2$ interior nodes, with $\mathbf{A}, \mathbf{B}$ constant tridiagonal matrices. Crank–Nicolson is second order in *both* time and space, $\mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta y^2)$.

Proposition 3 (Unconditional stability of Crank–Nicolson). *Scheme (13) is unconditionally L^2 -stable: $|\xi(\theta)| \leq 1$ for every θ and every $\Delta t > 0$.*

Proof. Let $g(\theta) = -4s_d \sin^2(\theta/2) - i s_a \sin \theta$ be the symbol of the spatial stencil, so that the explicit factor is $1 + g$ and the implicit one is $1 - g$. Crank–Nicolson amplifies a mode by

$$\xi(\theta) = \frac{1 + \frac{1}{2}g}{1 - \frac{1}{2}g}.$$

Write $g = -p - ir$ with $p = 4s_d \sin^2(\theta/2) \geq 0$ and $r = s_a \sin \theta$. Then

$$|\xi|^2 = \frac{(1 - \frac{p}{2})^2 + \frac{r^2}{4}}{(1 + \frac{p}{2})^2 + \frac{r^2}{4}} \leq 1,$$

because $p \geq 0$ implies $(1 + \frac{p}{2})^2 \geq (1 - \frac{p}{2})^2$. The bound holds for all θ and all Δt . $\square$

Because $\mathbf{A}$ is constant in time, we factor it once with a sparse LU decomposition and reuse the factors at every step, so each step costs only a back-substitution. The core of the implementation is shown in Listing 1.

Listing 1: Crank–Nicolson operators and time march (excerpt).

```

1 def _cn_operators(dy, nu, sig, dt, M):
2     sd = (0.5*sig**2)*dt/dy**2 # diffusion number
3     sa = nu*dt/dy # advection (Courant) number
4     lowL, diaL, uppL = -(sd/2+sa/4), 1+sd, -(sd/2-sa/4)
5     lowR, diaR, uppR = (sd/2+sa/4), 1-sd, (sd/2-sa/4)
6     A = diags([full(M-1,lowL), full(M,diaL), full(M-1,uppL)], [-1,0,1], "csc")
7     B = diags([full(M-1,lowR), full(M,diaR), full(M-1,uppR)], [-1,0,1], "csr")
8     return A, B
9
10 lu = splu(A) # factor once
11 for _ in range(nsteps): # reuse every step
12     q[1:-1] = lu.solve(B.dot(q[1:-1]))
13     q[0] = q[-1] = 0.0 # Dirichlet boundaries

```

4 Validation

All experiments use the base case $W_0 = 100$, $\mu = 0.08$, $\sigma = 0.18$, $T = 10$ years, giving the log-drift $\nu = \mu - \frac{1}{2}\sigma^2 = 0.0638$, a median terminal wealth $W_0 e^{\nu T} = 189.3$ and a mean $W_0 e^{\mu T} = 222.6$.

4.1 Agreement with the closed-form density

Figure 1 overlays the terminal wealth density produced by both schemes on the exact log-normal (7). The curves are visually indistinguishable. On a grid of $N = 1601$ points the Crank–Nicolson solution conserves total probability to $\int q \, dy = 1.000000$ and attains a maximum pointwise error of $\max_y |q_{\text{CN}} - q_{\text{exact}}| = 1.77 \times 10^{-5}$; the explicit scheme, run at the CFL-safe diffusion number $s_d = 0.20$, reaches 1.40×10^{-5} . The inset shows the pointwise error of both schemes is uniformly of order 10^{-5} across the support.

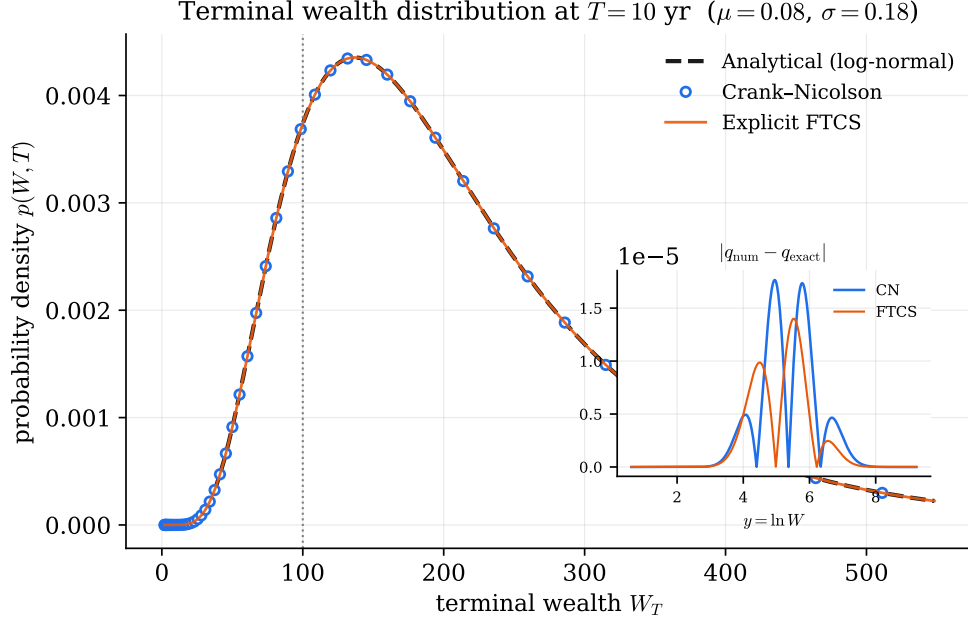


Figure 1: Terminal wealth density at $T = 10$ yr. Analytical log-normal (dashed), Crank–Nicolson (markers) and explicit FTCS (solid) coincide. Inset: pointwise error $|q_{\text{num}} - q_{\text{exact}}|$ on the log-grid, $\mathcal{O}(10^{-5})$ for both.

4.2 Convergence under grid refinement

To measure the *spatial* order of Crank–Nicolson we hold the time step fixed and small ($\Delta t = 2 \times 10^{-3}$, so the $\mathcal{O}(\Delta t^2)$ temporal error is negligible) and refine Δy . The explicit scheme cannot be treated this way—its stability forces $\Delta t \propto \Delta y^2$ —so for FTCS the temporal and spatial errors shrink together and the total is again $\mathcal{O}(\Delta y^2)$. Table 1 and Figure 2 report the max-norm error against the exact solution.

Table 1: Grid-refinement study: maximum-norm error against the exact Gaussian. Estimated orders are the slopes of $\log(\text{error})$ versus $\log(\Delta y)$.

Crank–Nicolson (fixed Δt)			Explicit FTCS ($\Delta t \propto \Delta y^2$)		
N	Δy	error	N	Δy	error
201	0.04303	1.11×10^{-3}	201	0.04303	1.25×10^{-3}
401	0.02152	2.78×10^{-4}	401	0.02152	2.16×10^{-4}
801	0.01076	6.94×10^{-5}	801	0.01076	5.62×10^{-5}
1601	0.00538	1.74×10^{-5}	1601	0.00538	1.35×10^{-5}
3201	0.00269	4.34×10^{-6}	—	—	—
observed order ≈ 2.00			observed order ≈ 2.15		

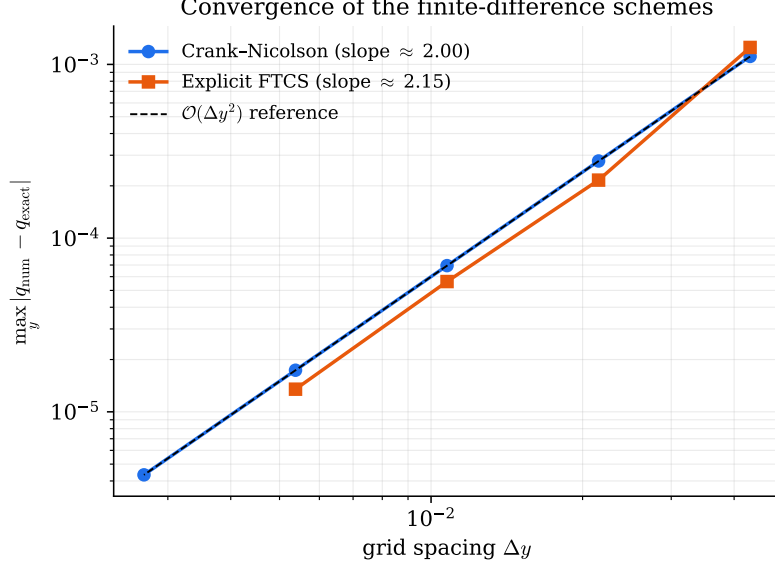


Figure 2: Max-norm error versus grid spacing on log–log axes. Both schemes track the $\mathcal{O}(\Delta y^2)$ reference line; the fitted slopes are 2.00 (Crank–Nicolson) and 2.15 (explicit FTCS).

Remark 2. Coupling $\Delta t \propto \Delta y$ for Crank–Nicolson—a common but careless choice—makes the temporal $\mathcal{O}(\Delta t^2)$ and spatial $\mathcal{O}(\Delta y^2)$ truncation terms equal in order. Because they carry opposite signs they cancel intermittently as the grid is refined, producing a non-monotone error and a misleadingly shallow apparent slope. Fixing Δt isolates the spatial order and recovers the clean second-order decay reported above.

4.3 Monte-Carlo cross-check

As an independent confirmation, Figure 3 simulates the SDE. Since log-wealth is exactly Gaussian, 2×10^5 terminal samples are drawn directly; the resulting histogram matches both the analytic density and the Crank–Nicolson solution. The left panel shows a sample of trajectories with the analytic median $W_0 e^{\nu t}$ and mean $W_0 e^{\mu t}$ overlaid, illustrating the widening, right-skewed cone of outcomes that the density in Figure 4 quantifies.

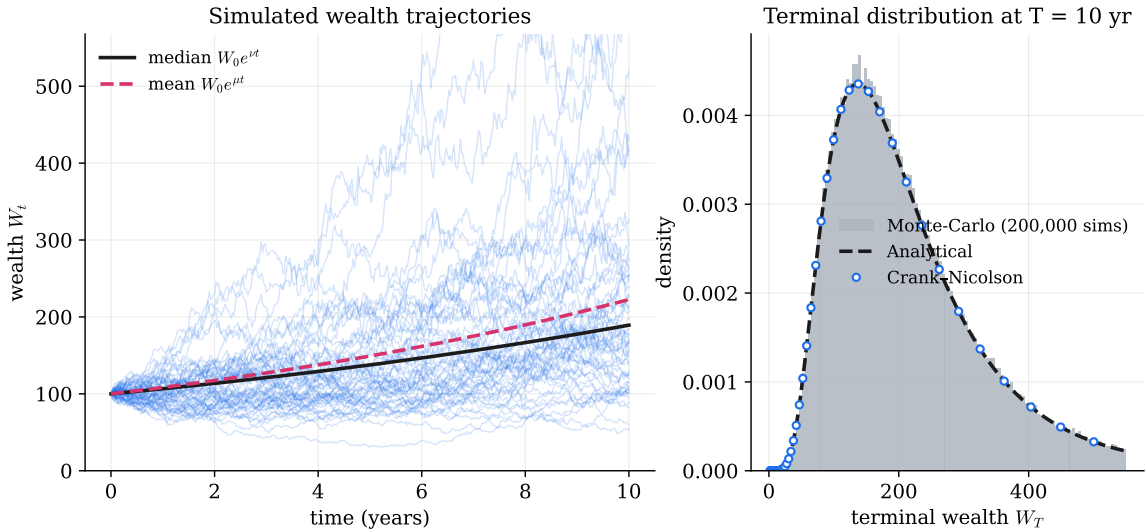


Figure 3: Left: simulated wealth trajectories with analytic median and mean. Right: terminal histogram from 2×10^5 simulations against the analytic density and the Crank–Nicolson solution.

5 Propagation of the wealth density

Figure 4 is the central qualitative result: snapshots of $p(W, t)$ recorded directly from the Crank–Nicolson march. At early times the density is a tall, narrow peak near $W_0 = 100$; as t grows it broadens and skews to the right, the signature of log-normal diffusion—multiplicative shocks compound, stretching the upper tail while the lower tail is bounded below by zero. By $T = 10$ the mode has drifted only modestly while the mean has moved well to the right, reflecting the gap between median (189) and mean (223) growth.

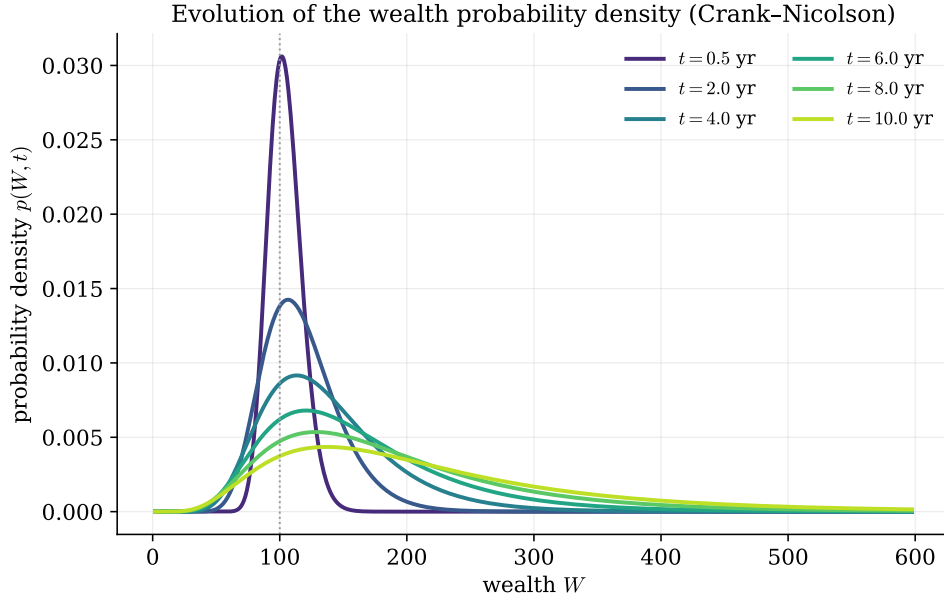


Figure 4: Evolution of the wealth probability density (Crank–Nicolson). An almost-point mass at W_0 spreads and develops the right skew characteristic of the log-normal law.

6 Risk metrics from the propagated density

Because the solver returns the *entire* density rather than only its moments, every risk measure follows by quadrature. We renormalise the terminal density (to absorb any $\mathcal{O}(10^{-6})$ truncation drift), form the cumulative distribution by trapezoidal integration, and read off the metrics of (8)–(9). Goal and floor are set to $G = 200$ (doubling) and $F = 80$ (a 20% drawdown). Table 2 compares the numbers obtained from the density to the closed form: they agree to four decimal places, closing the loop between the PDE solver and the analytic law.

Table 2: Risk metrics for the base case, computed from the Crank–Nicolson density by quadrature versus the closed-form log-normal expressions ($\alpha = 5\%$).

Metric	FDM (Crank–Nicolson)	Analytical
$\mathbb{P}(W_T \geq 200)$	0.4614	0.4614
$\mathbb{P}(W_T < 100)$	0.1312	0.1312
$\mathbb{P}(W_T \leq 80)$	0.0652	0.0652
5% VaR (wealth)	74.21	74.21
5% ES (wealth)	59.70	59.70
$\mathbb{E}[W_T]$	222.55	222.55

The reading is concrete: the base portfolio has a 46% chance of doubling over ten years and

a 13% chance of ending below its starting value; in the worst 5% of outcomes terminal wealth falls below 74.2, and the average of that tail is 59.7—a loss of roughly 40%.

7 Comparing asset allocations

The framework’s practical payoff is comparing portfolios on the same terms. A portfolio that holds a fraction w in equity (return $\mu_e = 0.08$, volatility $\sigma_e = 0.18$) and $1 - w$ in a risk-free asset (rate $r_f = 0.03$, zero volatility) has

$$\mu_p = w \mu_e + (1 - w) r_f, \quad \sigma_p = w \sigma_e, \quad (14)$$

and we solve (5) for each. Figure 5 shows the terminal densities of three archetypes; Table 3 lists their risk profiles.

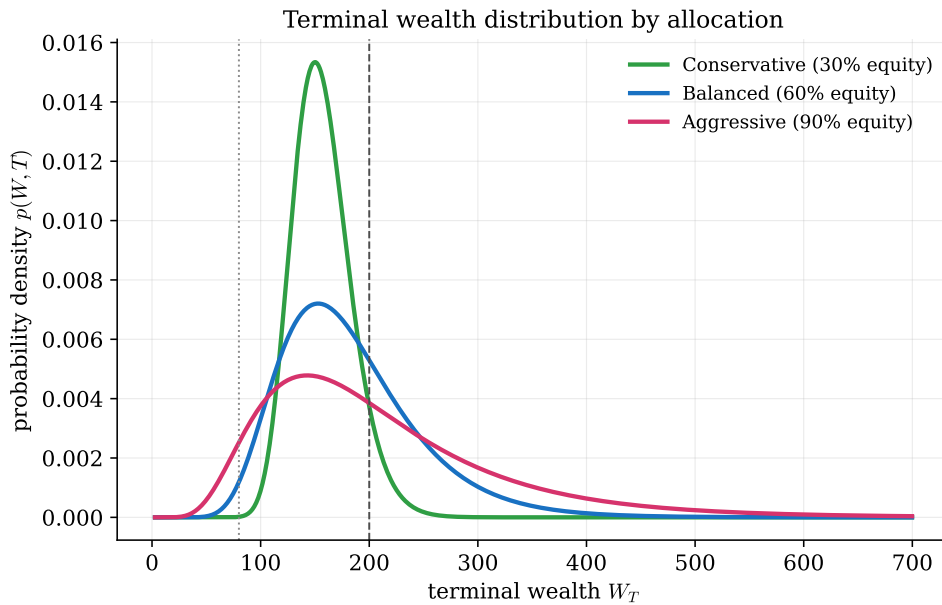


Figure 5: Terminal wealth densities for conservative (30% equity), balanced (60%) and aggressive (90%) allocations on a common grid. Dashed and dotted vertical lines mark the goal $G = 200$ and floor $F = 80$.

Table 3: Risk profiles by allocation, from the Crank–Nicolson terminal density.

Allocation	μ_p	σ_p	$\mathbb{P}(\text{reach } 200)$	$\mathbb{P}(\text{below } 100)$	5% VaR
Conservative (30%)	0.045	0.054	0.066	0.005	116.7
Balanced (60%)	0.060	0.108	0.329	0.056	98.0
Aggressive (90%)	0.075	0.162	0.442	0.114	79.9

The trade-off is stark and quantified. Moving from conservative to aggressive raises the probability of doubling from 7% to 44%, but the probability of ending below the initial capital climbs from 0.5% to 11%, and the 5% VaR slides from 116.7 (still a gain even in the bad tail) to 79.9 (a loss). The conservative portfolio almost never loses money but almost never reaches an ambitious goal; the aggressive one does the opposite.

Sweeping the equity weight continuously, Figure 6 traces these two probabilities and the 5th-percentile terminal wealth across $w \in [0, 1]$. The FDM markers lie on the analytic curves

throughout. The 5th-percentile wealth crosses the break-even line $W_T = W_0$ near $w \approx 0.63$: below that equity weight even a bad outcome preserves capital, above it the lower tail dips into losses.

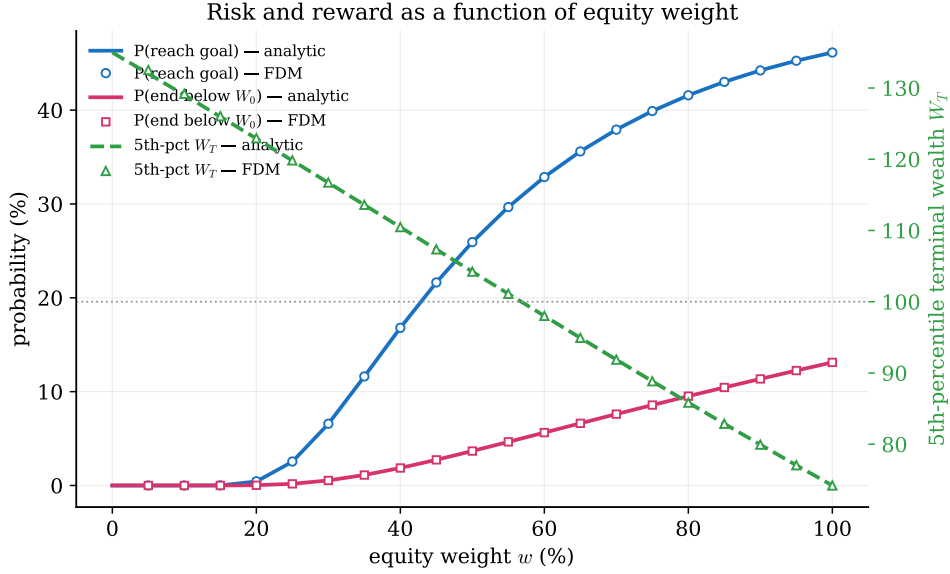


Figure 6: Risk and reward as a function of equity weight. Left axis: probability of reaching the goal and probability of ending below W_0 . Right axis: 5th-percentile terminal wealth. Solid/dashed lines are analytic; markers are the finite-difference solver.

8 Discussion, limitations and extensions

The results confirm that the Fokker–Planck approach reproduces the known log-normal law to high accuracy and yields the full battery of risk metrics by quadrature. Several caveats and natural extensions are worth stating.

What the closed form buys us, and why we still need the solver. For constant μ, σ the analytic density makes the PDE solver strictly unnecessary—its value here is as a *validated instrument*. The point of validation is that the same code, unchanged, applies where no closed form exists:

- **Local volatility** $\sigma = \sigma(W, t)$ and **state-dependent drift** $\mu = \mu(W, t)$: the coefficients in (3) simply become non-constant; the log-transform may no longer fully constant-ise them, but the finite-difference stencil is unchanged.
- **Mean-reverting wealth or returns** (e.g. an Ornstein–Uhlenbeck factor): introduces a linear restoring drift, still a parabolic PDE.
- **Consumption or contributions**: a deterministic cash-flow term adds a source/sink, i.e. an extra advection term, to (5).

Boundary treatment. Homogeneous Dirichlet conditions on a domain seven standard deviations wide truncate a mass of order 10^{-12} , well below the discretisation error, so they are harmless here. For heavier-tailed dynamics or longer horizons one would either widen the domain or impose zero-flux (Neumann) conditions $\partial_y q = 0$ to conserve probability exactly.

Dimensionality. A genuinely multi-asset portfolio with correlated risky holdings leads to a multi-dimensional Fokker–Planck equation. Finite differences then suffer the curse of dimensionality; beyond two or three factors, alternating-direction-implicit (ADI) splitting or a shift to Monte-Carlo or spectral methods becomes preferable. The one-dimensional reduction used here is exact only because the risk-free asset carries no volatility, so the portfolio remains a single diffusing state.

Model risk. GBM assumes constant volatility, continuous paths and independent log-returns—it excludes jumps, volatility clustering and fat tails. The probabilities in Table 2 are therefore conditional on the model; the methodological contribution is the machinery for turning *any* such diffusion model into goal and tail probabilities, not the specific numbers.

9 Conclusion

We have framed portfolio risk as the forward evolution of a probability density and solved the governing Fokker–Planck equation numerically. Reducing geometric Brownian motion to a constant-coefficient advection–diffusion equation by a logarithmic change of variables, we implemented explicit FTCS and Crank–Nicolson finite-difference schemes, proved their stability (conditional $s_d \leq \frac{1}{2}$ for FTCS, unconditional for Crank–Nicolson) and verified second-order spatial convergence. The solver reproduces the closed-form log-normal density to a maximum error of 1.8×10^{-5} , conserves probability to six digits, and agrees with Monte-Carlo simulation. From the propagated terminal density we recovered goal-attainment probabilities, Value-at-Risk and Expected Shortfall—matching the analytic formulas to four decimals—and used them to rank conservative, balanced and aggressive allocations, making the risk–reward trade-off explicit and quantitative. Because the method never uses the closed form in its computation, it transfers directly to the state- and time-dependent models where analytic densities are unavailable, which is exactly where a PDE solver earns its place.

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